Metric
DOD-I-63276/1A
21 October 1983
SUPERSEDING
DOD-I-63276/1
26 September 1977

MILITARY SPECIFICATION SHEET

INSERT, SCREW THREAD, LOCKED IN, RING LOCKED, SERRATED, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the insert described herein shall consist of this specification and the latest issue of specification DOD-I-63276.

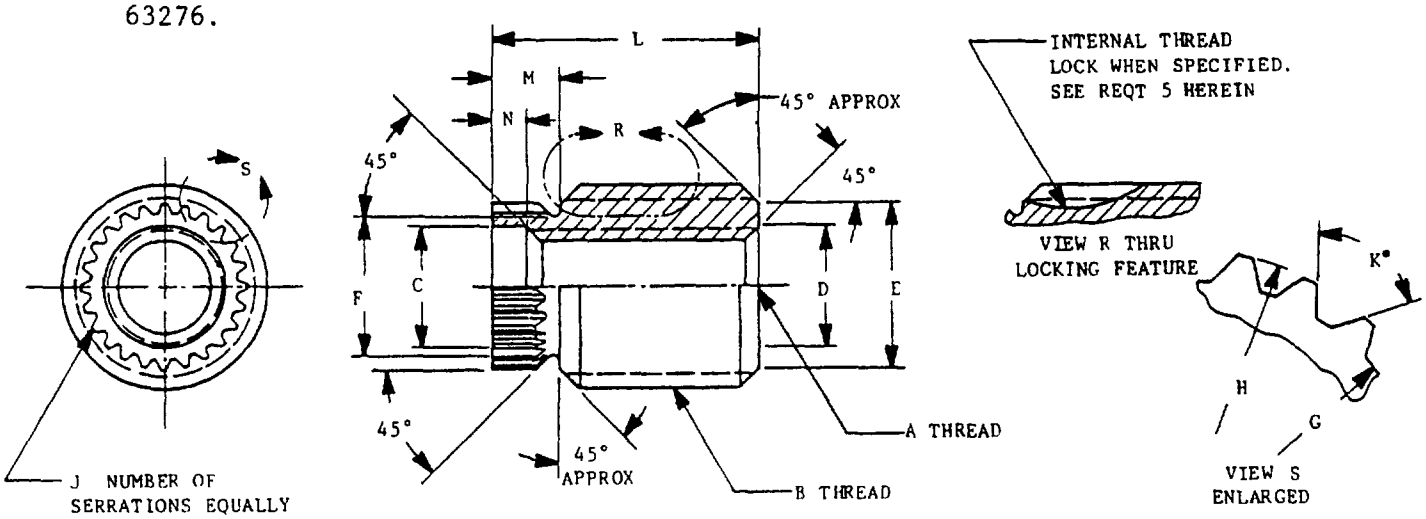


FIGURE 1. Insert, screw thread, details

NOTES:

1. Unless otherwise specified, dimensions are in millimeters, and shall apply after plating and before the addition of solid film lubricant.
2. Fillets shall be 0.8 R maximum.
3. Edges shall be broken 0.1-0.4 unless otherwise specified.

TABLE I. Part dimensions and dash numbers, short length

DASH NUMBER	A INTERNAL THREAD	B EXTERNAL THREAD	C DIA +0.15 -0.00	D DIA ±0.25	E DIA ±0.20	F DIA MIN	G DIA MAX	H DIA +0.00 -0.20	J	K° +2° -1°	L ±0.45	M ±0.3	N MAX	MASS kg/1000 PCS APPROX 1/	BASIC LOCKRING SIZE
101	MJ5x0.8	M10x1 5	5.20	5.40	7.73	5.65	6.57	7.40	17	86°	10.50	3.4	1.6	2.75	DOD-I-63276/2-2
102	MJ6x1	M12x1.5	6.20	6.44	9 73	7.57	8.51	9.45	17	99°	11.60	4.0	2.0	4.90	DOD-I-63276/2-3
103	MJ7x1	M14x1.5	7.20	7.44	11.73	9.48	10.59	11.61	21	97°	13.80	4.6	2.0	8 65	DOD-I-63276/2-4
104	MJ8x1.25	M14x1.5	8.20	8.48	11 73	9 48	10.59	11.61	21	97°	15 90	4 6	2 5	9.05	DOD-I-63276/2-4
105	MJ8x1														
106	MJ10x1 5	M16x1 5	10.20	10 53	13 73	11 48	12.22	13.21	24	96°	19.60	5 6	3 0	13.70	DOD-I-63276/2-5
107	MJ10x1.25														
108	MJ12x1.5	M20x1.5	12.20	12.53	17 73	13.48	14.48	15.45	26	111°	21.90	6.0	3.0	25.40	DOD-I-63276/2-6
109	MJ12x1.25														
110	MJ14x1.5	M22x1 5	14.20	14.53	19 73	15.48	17.16	18 47	28	96°	24 90	6.0	3 0	34.60	DOD-I-63276/2-7

1/ Calculated with 7.9 kg/mm^3

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TABLE II. Part dimensions and dash numbers, medium length

DASH NUMBER	A INTERNAL THREAD	B EXTERNAL THREAD	C DIA +0.15 -0.00	D DIA ±0.25	E DIA ±0.20	F DIA MIN	G DIA MAX	H DIA +0.00 -0.20	J	K° +2° -1°	L ±0.45	M ±0.3	N MAX	MASS kg/1000 PCS APPROX 1/	BASIC LOCKRING SIZE
201	MJ5x0.8	M10x1.5	5.20	5.40	7.73	5.65	6.57	7.40	17	86°	11.70	3.4	1.6	3.20	DOD-I-63276/2-2
202	MJ6x1	M12x1.5	6.20	6.44	9.73	7.57	8.51	9.45	17	99°	13.00	4.0	2.0	5.65	DOD-I-63276/2-3
203	MJ7x1	M14x1.5	7.20	7.44	11.73	9.48	10.59	11.61	21	97°	15.50	4.6	2.0	10.00	DOD-I-63276/2-4
204	MJ8x1.25	M14x1.5	8.20	8.48	11.73	9.48	10.59	11.61	21	97°	18.20	4.6	2.5	10.70	DOD-I-63276/2-4
205	MJ8x1														
206	MJ10x1.5	M16x1.5	10.20	10.53	13.73	11.48	12.22	13.21	24	96°	22.80	5.6	3.0	16.40	DOD-I-63276/2-5
207	MJ10x1.25														
208	MJ12x1.5	M20x1.5	12.20	12.53	17.73	13.48	14.48	15.45	26	111°	25.70	6.0	3.0	31.00	DOD-I-63276/2-6
209	MJ12x1.25														
210	MJ14x1 5	M22x1 5	14.20	14.53	19.73	15.48	17.16	18.47	28	96°	29.60	6.0	3.0	47.00	DOD-I-63276/2-7

1/ Calculated with 7.9 kg/mm^3

TABLE III. Part dimensions and dash numbers, long length

DASH NUMBER	A INTERNAL THREAD	B EXTERNAL THREAD	C DIA +0.15 -0.00	D DIA ±0.25	E DIA ±0.20	F DIA MIN	G DIA MAX	H DIA +0.00 -0.20	J	K° +2° -1°	L ±0.45	M ±0.3	N MAX	MASS kg/1000 PCS APPROX 1/	BASIC LOCKRING SIZE
301	MJ5x0.8	M10x1.5	5.20	5.40	7.73	5.65	6.57	7.40	17	86°	13.10	3.4	1.6	3.70	DOD-I-63276/2-2
302	MJ6x1	M12x1.5	6.20	6.44	9.73	7.57	8.51	9.45	17	99°	14.60	4.0	2.0	6.55	DOD-I-63276/2-3
303	MJ7x1	M14x1.5	7.20	7.44	11.73	9.48	10.59	11.61	21	97°	17.50	4.6	2.0	11.60	DOD-I-63276/2-4
304	MJ8x1.25	M14x1.5	8.20	8.48	11.73	9.48	10.59	11.61	21	97°	20.90	4.6	2.5	12.60	DOD-I-63276/2-4
305	MJ8x1														
306	MJ10x1.5	M16x1.5	10.20	10.53	13.73	11.48	12.22	13.21	24	96°	26.50	5.6	3.0	19.60	DOD-I-63276/2-5
307	MJ10x1.25														
308	MJ12x1.5	M20x1.5	12.20	12.53	17.73	13.48	14.48	15.45	26	111°	30.20	6.0	3.0	37.40	DOD-I-63276/2-6
309	MJ12x1.25														
310	MJ14x1.5	M22x1.5	14.20	14.53	19.73	15.48	17.16	18.47	28	96°	35.10	6.0	3.0	51.50	DOD-I-63276/2-7

1/ Calculated with 7.9 kg/mm^3

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REQUIREMENTS:

1. Material: Steel, alloy, composition 8740 (UNS G87400) per AMS6322. Steel, corrosion-resistant, composition 17-4PH (UNS S17400) per AMS5643. Steel, corrosion and heat resistant, composition A-286 (UNS K66286) per AMS5731 or AMS5734. See material code letters per Requirement 4.

2. Protective coating or treatment: Alloy steel (UNS G87400) shall be cadmium plated in accordance with general specification. Corrosion resistant steel (UNS S17400 locked parts shall be coated with solid film lubricant and non-locked parts shall be cleaned, descaled and passivated in accordance with general specification. Corrosion and heat resistant steel (UNS K66286), locked parts shall be silver plated and non-locked parts shall be cleaned, descaled and passivated in accordance with general specification.

3. Surface texture: 3.2µm except serrations.

4. Mechanical properties:

Material Code

Letter	Material	Hardness, Minimum
A	Steel, Alloy Comp. 8740	34HRC
B	Steel, Corrosion Resistant Comp. 17-4PH	321HB
C	Steel, Corrosion and Heat Resistant, Comp. A-286	248HB

5. Internal thread lock: When internal thread lock feature is specified by part number letter "L", material "A" shall be cadmium plated, material "B" shall be coated with solid film lubricant, and material "C" shall be silver plated in accordance with Requirement 2 herein.

6. Military part number: Consists of the letter "D", plus the basic number of the specification sheet, plus the material code, plus the applicable dash number from tables I, II, or III, plus letter "L" for internal thread lock feature.

Example of Military part number:

D63276/1-	C	305
Basic number of specification sheet	Material Code	Dash No.

Explanation of part number example:

Insert, Steel, CRES, A-286, long length, MJ8x1 internal thread size, with no internal thread locking feature.

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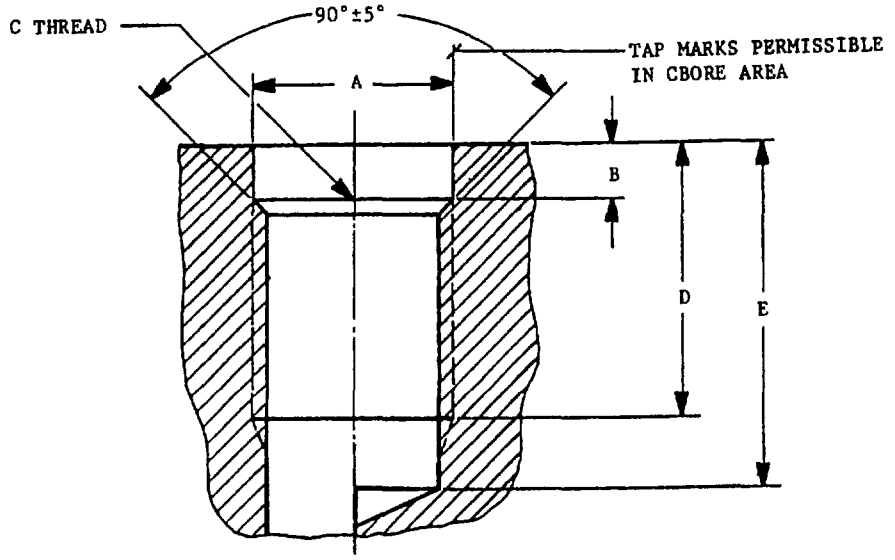


FIGURE 2. Parent material hole preparation

TABLE IV. Installation criteria in parent material

INTERNAL THREAD SIZE OF INSERT REF	INSERT DASH NUMBER REF	A DIA	B +0.4 -0.0	C THREAD TOLERANCE CLASS-4H5H	D FULL THD DEPTH MIN	E DRILL DEPTH BLIND HOLE MIN
MJ5x0.8	101	10.00-10.13	3.0	M10x1.5	11.4	15.0
	201				12.6	16.2
	301				14.0	17.6
MJ6x1	102	12.00-12.13	4.0	M12x1.5	12.5	16.1
	202				13.9	17.5
	302				15.5	19.1
MJ7x1	103	14 00-14.15	4 7	M14x1.5	14.7	18.3
	203				16.4	20.0
	303				18.4	22.0
MJ8x1.25	104	14.00-14.15	4 7	M14x1.5	16.8	20.4
	204				19.1	22.7
	304				21.8	25.4
MJ8x1	105	14.00-14.15	4.7	M14x1.5	16.8	20.4
	205				19.1	22.7
	305				21.8	25.4
MJ10x1.5	106	16.00-16.15	5.7	M16x1.5	20.5	24.1
	206				23.7	27.3
	306				27.4	31.0
MJ10x1.25	107	16.00-16.15	5.7	M16x1.5	20.5	24.1
	207				23.7	27.3
	307				27.4	31.0
MJ12x1.5	108	20.00-20.18	6.0	M20x1.5	22.8	26.4
	208				26.6	30.2
	308				31.1	34.7
MJ12x1.25	109	20.00-20.18	6.0	M20x1.5	22.8	26.4
	209				26.6	30.2
	309				31.1	34.7
MJ14x1.5	110	22.00-22.18	6.0	M22x1.5	25.8	29.4
	210				30.5	34.1
	310				36.0	39.6

NOTES:

1. Diameter "A" and thread shall be concentric within 0.1 FIM.
2. Axis of hole shall be normal to entry surface or provide spotface when required.
3. Machined surfaces shall be 3.2µm in accordance with ANSI B46.1.
4. Dimensions are in millimeters.

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5. Insert installation procedure.
5.1 Install insert to depth shown in figure 3.
5.2 Drive lockring to depth shown in figure 4.

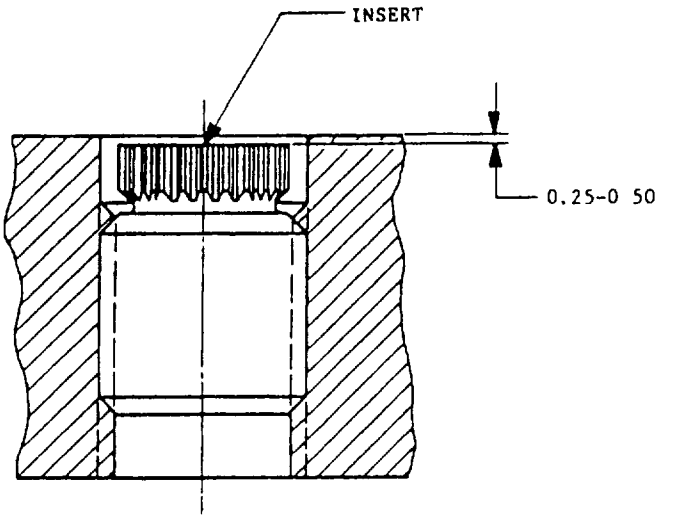


FIGURE 3 Insert installed

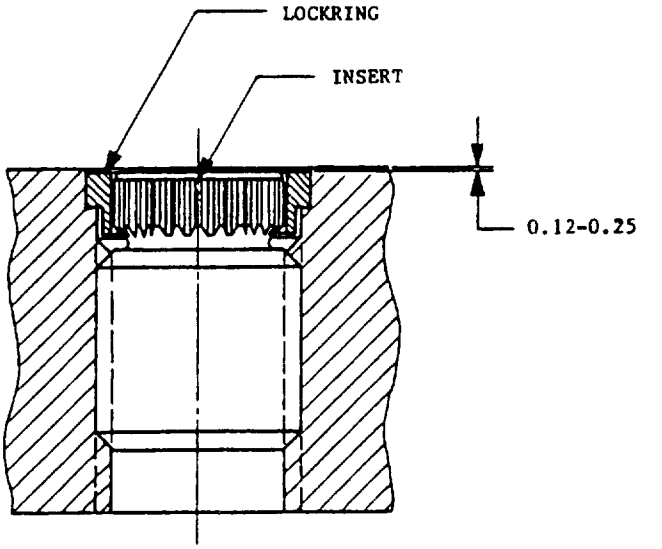


FIGURE 4 Completed installation

5.3 Replacement of inserts is made with same size parts as those removed and in same manner as originally installed. Rotate new lockring if necessary, before driving, to a position in which the external serrations are aligned with those in parent material.

5.4 Use of manufacturers installation tools mandatory (Rosan, Inc. Newport Beach, CA - FSCM 83324).

Custodians:
Army - AR
Navy - OS
Air Force - 99

Review activities:
Army - AV
Air Force - 11
NSA - NS
DLA - IS

User activities:
Army - AT, ME
Navy - MC

Preparing activity:
Army - AR

(Project 5340-1464)

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